FitBuddy

Solution Requirements

Purpose: define what the first version of the AI fitness plan generator must do.

1. Functional requirements

FR1. The system shall let a user enter name, age, weight, fitness goal, and workout intensity.
FR2. The system shall validate required fields and accepted values before processing.
FR3. The system shall generate a structured 7-day workout plan using the submitted details.
FR4. Each plan should include daily activities and suitable warm-up/recovery guidance.
FR5. The system shall provide a concise nutrition or recovery tip related to the goal.
FR6. The user shall be able to submit feedback to request a plan change.
FR7. The system shall save user details and the original and updated plan versions.
FR8. The system shall show the generated or updated plan in a readable results page.
FR9. An authorized coach/admin shall be able to view users and their plan versions.

2. Data requirements

Store a user ID, name, age, weight, goal, intensity, original plan, feedback, updated plan, and relevant timestamps. Use a consistent user ID and preserve the original plan.

3. Interface requirements

Provide a simple input page, readable results page, feedback form, and coach/admin view. Label fields clearly, show validation errors, and support common desktop and mobile screens.

4. Security and privacy requirements

Protect the Gemini API key in environment configuration; never expose it in pages or source control. Restrict the coach/admin view to authorized users. Collect only necessary personal data, explain its use, and protect stored records from unauthorized access.

5. Quality and reliability requirements

Show a clear message if AI generation fails; avoid losing submitted data where possible. Keep outputs organized and easy to scan. Handle invalid input and missing records safely. Use safe, general fitness language; invite users to adapt activities to their abilities.

Acceptance checks

A valid submission produces and displays a 7-day plan and tip; invalid fields show guidance. Feedback produces a revised plan while retaining the original; an authorized admin can review saved records. The API key is not visible in the browser or public repository.

Derived from the supplied FitBuddy brief. These requirements describe a proposed first release.